



Waves in Fluids

Chapter 9

In this chapter, we will explain wave propagation in fluids. When considering wave propagation in the atmosphere and ocean, the restoring force due to gravity plays an essential role. As explained in Chapter ??, gravity is naturally viewed from an Eulerian perspective, while stress is essentially naturally viewed from a Lagrangian perspective. Thus care must be taken when analyzing it. Therefore, in this chapter, we will first consider a zero-order approximate picture of waves in the atmosphere and ocean, and then explain their behavior based on the governing equations in the latter half.

§9.1 Boundary Waves: Lamb Waves and Ocean Gravity Waves

When considering atmospheric and oceanic waves, the Earth's surface must be treated as a different boundary condition. This is because the atmosphere is a gas and its density is much smaller than that of the ocean. Therefore, for the atmosphere, the surface can be approximated as a fixed boundary (rigid wall), whereas for the ocean, the surface can be approximated as a free boundary. First, as a simple example, we will explain the physical picture of atmospheric Lamb waves and oceanic gravity waves, which are boundary waves with amplitudes near the surface, in a simplified manner.

9.1.1 Lamb Waves

Considering sound waves in the ocean, do horizontally propagating sound waves exist near the water surface? As explained in Section 5.6.1, if the presence of the atmosphere can be neglected and the water surface can be regarded as a free boundary surface, horizontally propagating sound waves cannot exist. How about considering sound waves in the atmosphere?

For the atmosphere, the surface of the Earth can be approximated as a rigid wall. Therefore, we can see that we need to consider the boundary condition where the vertical displacement is 0. In other words, sound waves that repeatedly compress and expand only in the horizontal direction can exist. These are known as atmospheric Lamb waves. As we will explain in detail later, strictly speaking, atmospheric Lamb waves arrive at a hydrostatic equilibrium in the vertical direction, and propagate at the speed of sound while repeatedly compressing and expanding in the horizontal direction.

It is known that when a large explosion occurs in the atmosphere, Lamb waves are actually generated and orbit the Earth. Recently, it was observed during the 2022 eruption of the Hunga Tonga-Hunga Ha'apai volcano (Figure 9.1). It has been reported that they were recorded by barometers worldwide during atmospheric nuclear tests along with the 1960s.⁽⁷⁾ Also, it is known that atmospheric Lamb waves are steadily excited not only during volcanic eruptions but also continuously by meteorological phenomena.⁽⁸⁾

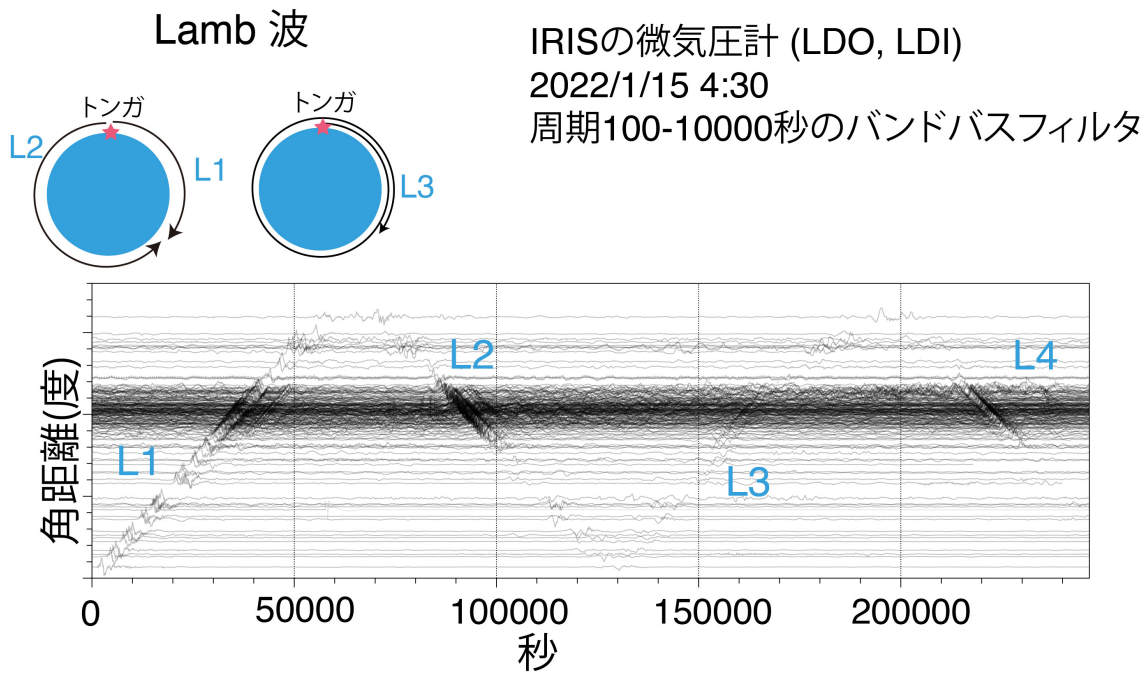


Figure 9.1: Barometer records when the Hunga Tonga-Hunga Ha'apai volcano erupted in 2022. The horizontal axis corresponds to a period of about 3 days, and the vertical axis corresponds to angular distances from 0° to 90°. You can see Lamb waves orbiting the Earth with a propagation speed of about 300 m/s.

9.1.2 Oceanic Gravity Waves

From the ocean's point of view, the sea surface behaves as a free end. As stated earlier, when considering sound waves propagating in a semi-infinite medium, horizontally propagating sound waves cannot exist due to the effect of the free surface. However, gravity waves, whose restoring force is gravity, exist in the ocean. First, let us build a physical picture of their behavior.

Shallow Water Waves

Oceanic gravity waves are called shallow water waves when their wavelength is longer than the water depth. Tsunamis are an example. Here let's roughly consider the physics behind the propagation.

Let the horizontal flow velocity be $i\omega s_x$ and the vertical flow velocity be $i\omega s_z$. When the wavelength is sufficiently long compared to the water depth, the flow can be considered two-dimensional, and the horizontal flow velocity becomes constant relative to the vertical direction. Assuming the water depth is H , considering the conservation of mass, we obtain

$$i\omega \frac{ds_x}{dx} H + i\omega \eta = 0 \quad (9.1)$$

where η represents the surface elevation.

The horizontal flow dominates and the pressure gradient generated by the sea surface fluctuation drives the flow. Therefore, the equation of motion becomes

$$-\rho\omega^2 s_x = -\frac{\partial \rho g \eta}{\partial x} \quad (9.2)$$

Eliminating s_x , we get

$$-\omega^2 \eta = gH \frac{\partial^2 \eta}{\partial x^2} \quad (9.3)$$

which indicates that the propagation velocity of shallow water waves is $\sqrt{gH}$.

Deep Water Waves

Although it may seem somewhat top-down, let us try to construct an intuitive physical picture for the case where the wavelength is sufficiently short compared to the water depth.

When considering a wave with angular frequency ω , the physical quantities that determine the system are only g and ρ . If we try to determine the propagation velocity from dimensional analysis, we find that the combination yielding velocity is only g/ω . It is expected that the propagation velocity becomes slower as the frequency increases. This is because the restoring force due to gravity works two-dimensionally near the water surface, whereas the mass moved by the wave is proportional to the cube of its wavelength.

Let's consider how the conservation of mass and the equation of motion balance. When the wavelength is shorter than the depth, the resulting flow is limited to a depth of roughly the wavelength near the water surface. Letting the wavelength be λ , the conservation of mass implies

$$i\omega \frac{ds_x}{dx} \lambda \propto -i\omega \eta \quad (9.4)$$

Recalling the relation that the wavelength $\lambda = 2\pi/k$, we have

$$\frac{ds_x}{dx} \propto -k\eta \quad (9.5)$$

Considering the flow directly beneath the sea surface, it is driven by the pressure gradient caused by fluctuations in the surface elevation. Therefore, the equation of motion reads

$$-\rho\omega^2 s_x = -\frac{\partial \rho g \eta}{\partial x} \quad (9.6)$$

Eliminating s_x , we have

$$-\omega^2 \eta \propto \frac{g}{k} \frac{\partial^2 \eta}{\partial x^2} \quad (9.7)$$

In other words, the phase velocity of deep water waves is proportional to g/ω . Since this is an order-of-magnitude estimation, the proportionality coefficient cannot be evaluated. However, as derived later, we'll see that the proportionality coefficient is exactly 1. Because the group velocity is given by $\frac{d\omega}{dk}$, it can be deduced to be $g/(2\omega)$.

As an example, let us look at ocean bottom pressure gauge records during the 2022 eruption of the Hunga Tonga volcano. Figure 9.2 shows the running spectrum of the ocean bottom pressure gauge records observed in the Tohoku region. The vertical axis represents frequency, and the horizontal axis represents time. We can observe that the arrival time is linearly delayed in proportion to the frequency. Because the arrival time can be calculated by dividing the epicentral distance by the group velocity, we understand that it illustrates the dispersion characteristic of deep water waves.

Problem 9.1

1. From Figure 9.2, one can estimate the distance from the observation station to the Hunga volcano. Read the numerical values from the figure and evaluate that distance.

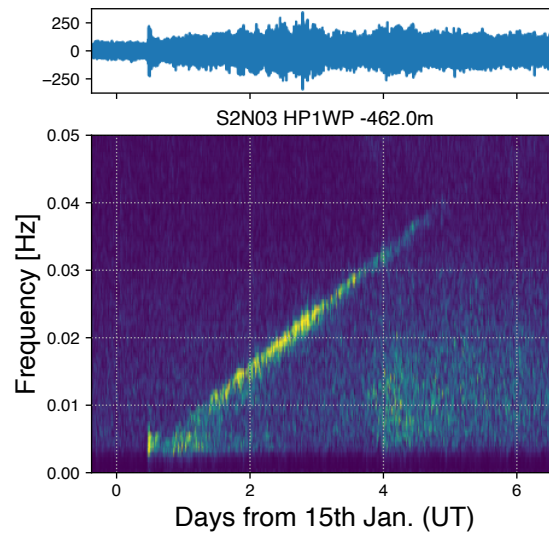


Figure 9.2: Oceanic gravity waves excited when the Hunga Tonga volcano erupted in 2022. It can be seen that the arrival time is delayed as the frequency becomes higher. This indicates that the group velocity is inversely proportional to frequency. We used ocean bottom pressure gauges from the S-net observation stations.

9.1.3 Wake Waves

In the introduction, we used the waves created by ducks on the water surface as an example to explain dispersion. Let us consider this in more detail here. The waves created by ducks are known as wake waves. For example, assume a ship is moving to the right at 20 m/s. Assuming simply that the ship is generating concentric waves at each time step, one can represent the wake wave by integrating the two-dimensional Green's function along the movement of the ship. Figure 9.3 reveals an actual calculation result. We can observe a pattern similar to the waves originated by ducks (known as the Kelvin wave pattern). Furthermore, if one were to observe the V-shaped waves at a single point in space, we recognize that higher-frequency components arrive later, reflecting the effects of dispersion.

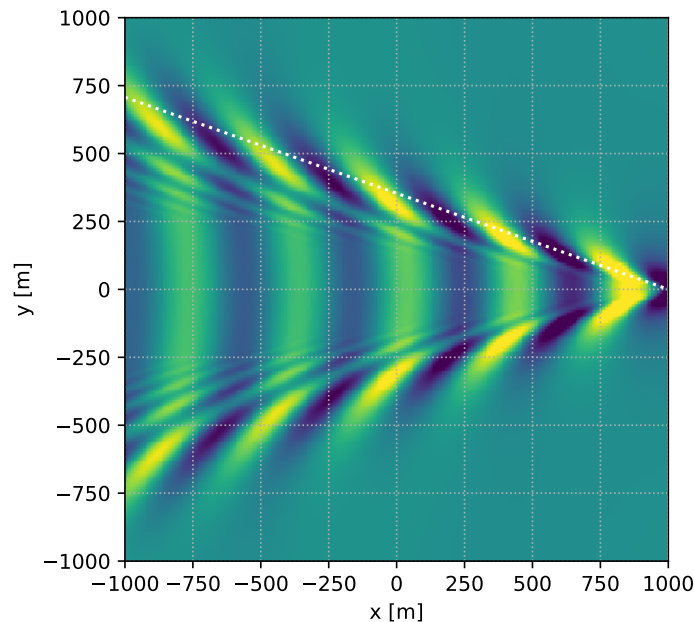


Figure 9.3: An example of a wake wave. The ship moves at 20 m/s and the upper limit of frequency is 0.25 Hz.

First, let us examine the phase velocity at frequency ω . For simplicity, assume there is no dispersion. Let the ship be moving toward the right at speed V in Figure 9.4(a). Similar to when analyzing the forerunner wave in Section 6.4.4, let us reflect on Huygens' principle. Two surfaces of equal phase extend backwards from the ship. This is known as a Mach cone (or shock cone). Assuming the angle is denoted by θ as in (b), the relation $\sin \theta = c_p/V$ is held between the phase velocity c_p and the ship speed V .

In the case of deep water waves, the problem becomes more complex because the group velocity is $1/2$ of the phase velocity. As shown in Figure 9.4(b), because the wave group progresses only $1/2$ relatively to the phase, the amplitude enlarges along the thick red line depicted in the figure. Let the included angle of this line be α . Since the phase itself is parallel to the constant-phase plane,

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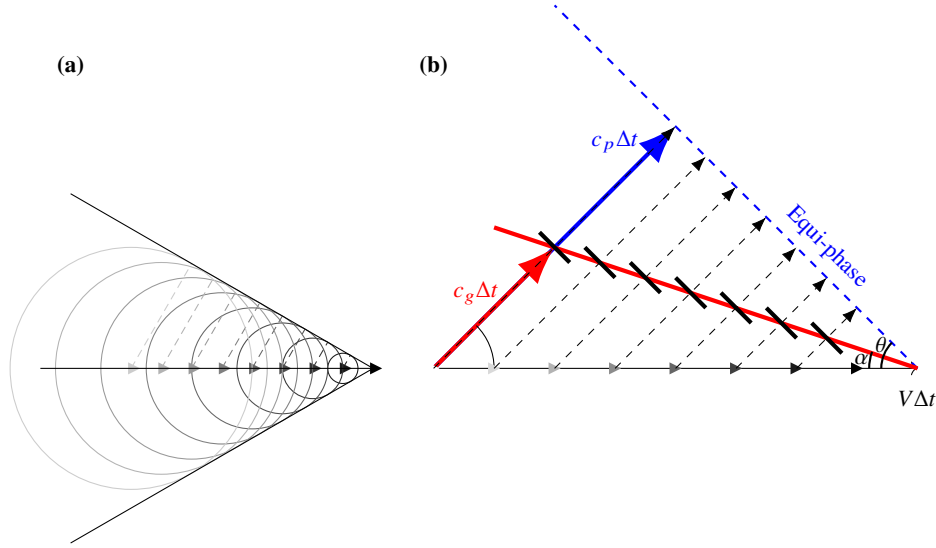


Figure 9.4: (a) Case with no dispersion. It corresponds to the shock cone formed by shock waves. (b) Schematic diagram of wake wave generation. It is vital to understand the relationship between the group velocity and phase velocity.

slanted lines align along the thick red line as shown in Figure (b). The phase velocity ($c_p = g/\omega$) and the group velocity ($c_g = g/(2\omega)$) are dependent on the frequency, indicating the angles α and θ depend strictly on frequency.

Now, let us examine the relationship between the angles α and θ . Through geometric interactions and trigonometric addition formulas, α and θ relate as follows:

$$\tan \alpha = \frac{1}{2} \frac{\tan \theta}{1 + \frac{\tan^2 \theta}{2}} \quad (9.8)$$

By differentiating with respect to $\tan \theta$ as a variable, we find that $\tan \alpha$ reaches its local maximum precisely at $\tan \alpha = \sqrt{2}/4$. Recalling the method of stationary phase (Section 6.4.5), it is expected that contributions near the local maximum of $\tan \alpha$ are large. This occurs because wave components of various wavelengths overlap in phase, consequently enhancing the amplitude. Regardless of the boat's speed, waves exhibit huge amplitudes at a constant angle α . Simply put, assuming the boat radiates waves equally at all frequencies, the angle α formed by the wave group is independent of frequency and equates to $\arctan(\sqrt{2}/4) \sim 19^\circ$.

In Figure 9.3, not only the wave groups aligned at angle α can be seen, but also wavefronts orthogonal to the propagating direction. Let us think about these low-frequency groups of waves. Assuming the speed of the boat is unchanged, the regions on the left of the peak correspond to slower waves in Figure 9.5, while the right side corresponds to faster waves. Simply put, the portion to the left of the maximum corresponds to the wavefronts currently under theoretical consideration. The equi-phase surfaces are fundamentally orthogonal to the advancing direction,

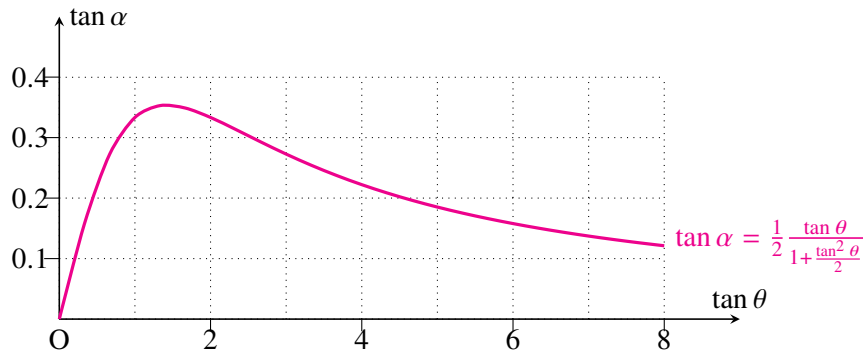


Figure 9.5: Relationship between the angle θ formed by equal-phase surfaces and angle α of the wave group. Fixing the speed of the ship, values right of the local maximum refer to faster waves, while left refers to slower waves. Because we derive with the angle θ , the wave propagation speed is constantly slower than the moving speed of the ship. Hence, a low-frequency cutoff establishes automatically from $g/2\omega = V$.

while the wave groups tend to be concentrated close to the x -axis. It suggests that a wavefront progresses horizontally along the x -axis at a very low frequency, remaining perpendicular to the direction. Clearly, this is the component we detected in Figure 9.3.

Problem 9.2

1. Derive equation 9.8.
2. Assuming the ship propagates at 20 m/s and radiates equally across all frequencies. Deduce the dominant frequency operating as a wave group.
3. Providing a figure mirroring Figure 9.4(b) as θ approaches 90° , elaborate on the characteristics of the resulting wave groups.

§9.2 Fluid motion incorporating the effect of gravity

Here we will consider the equations that describe wave propagation in the atmosphere and ocean.

When considering wave propagation in the atmosphere and ocean, the effect of gravity plays an important role compared to the propagation of seismic waves in solid bodies. The effects of

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gravity can be broadly divided into two. Here let us look at each effect ^{note 1)}.

The first is hydrostatic pressure. When assuming hydrostatic equilibrium is reached, the vertical pressure gradient and gravity balance each other. In the case of the atmosphere, pressure and density decrease exponentially, and in the case of the ocean, hydrostatic pressure increases proportionally to the water depth. When considering the initial stress that balances gravity in a steady state, care is required because the Eulerian perspective differs from the Lagrangian perspective.

The second effect is its role as a restoring force. As mentioned earlier, the atmosphere is density-stratified (becomes progressively thinner with altitude) due to gravity. Therefore, when oscillating vertically, buoyancy becomes an important restoring force. In this case, we need to consider not only sound waves but also gravity waves with gravity as the restoring force. It is easy to envision waves forming on the interface between oil and water in a dressing.

Assuming the time constant we are focusing on is sufficiently shorter than the rotation period, effects of Earth's rotation such as centrifugal and Coriolis forces are neglected. Also, for simplicity, advection effects such as mean winds are assumed to be negligible.

In the low-frequency band, atmospheric gravity waves, which have gravity as a restoring force, play an important role. When considering atmospheric gravity waves, the change in the gravitational potential caused by density changes can be approximated as being sufficiently smaller than buoyancy. Therefore, the gravitational potential here is neglected as being sufficiently smaller than the gravitational acceleration g_0 , but the effect of g_0 is taken into account (the Boussinesq approximation? The Cowling approximation ignores self-gravity) ^{(1), (3)note 2)}.

Let us summarize this into the form of differential equations with the vertical displacement U and stress T_{zz} as independent variables. We assume homogeneity in the horizontal direction. The complication arises because everything except stress takes Eulerian perturbations, whereas stress is Lagrangian. This is because the stress caused by deformation (Lagrangian) is added to the initial stress prior to deformation. Let the position before deformation be $\mathbf{x}$, the position after deformation be $\mathbf{r}$, and the displacement be $\mathbf{s} \equiv (U, V)^T = \mathbf{r} - \mathbf{x}$. The stress at position $\mathbf{r}$ is $-p_0(\mathbf{x}) + T_{zz}$. The hydrostatic pressure $p_0(\mathbf{r})$ at position $\mathbf{r}$ is $\approx p_0(\mathbf{x}) - \rho_0 g_0 U$. Hence, the Eulerian stress perturbation at position $\mathbf{r}$ is $T_{zz} + \rho_0 g_0 U$.

Equation of motion

$$-\rho_0 \omega^2 V = \frac{d(T_{zz} + \rho_0 g_0 U)}{dx} \quad (9.9)$$

$$-\rho_0 \omega^2 U = \frac{d(T_{zz} + \rho_0 g_0 U)}{dz} - g_0 \nabla \cdot (\rho_0 \mathbf{s}) \quad (9.10)$$

The last term in the second equation is the result of substituting the mass conservation equation (Section ??) into the buoyancy term.

^{note 1)}For details, refer to Chapter 6.14 Adjustment to Equilibrium in a Stratified Compressible Fluid of Gill's textbook.⁽⁵⁾ For a more detailed understanding of atmospheric sound waves, the out-of-print textbook by Gossard⁽⁶⁾ is a good resource.

^{note 2)}When calculating seismic waves, it is also possible to incorporate a fluid layer to account for the effects of the outer core or the ocean.^{(4), (9), (10)}

From the horizontal component

$$V = -\frac{ik}{\rho_0\omega^2}(T_{zz} + \rho_0 g_0 U) \quad (9.11)$$

From the vertical component

$$-\rho_0\omega^2 U = \frac{d(T_{zz} + \rho_0 g_0 U)}{dz} - \nabla \cdot (\rho_0 s) g_0 \quad (9.12)$$

$$= \frac{d(T_{zz} + \rho_0 g_0 U)}{dz} - \left(ik g_0 \rho_0 V + \frac{d(\rho_0 U)}{dz} \right) \quad (9.13)$$

$$= \frac{dT_{zz}}{dz} - ik g_0 \rho_0 V \text{ substituting horizontal component equation}$$

$$= \frac{dT_{zz}}{dz} + \frac{k^2 g_0}{\omega^2} (T_{zz} - \rho_0 g_0 U) \quad (9.14)$$

Assuming adiabatic deformation, from the equation of state (which must be considered in a Lagrangian manner: Section ??),

$$-\rho_0 \nabla \cdot (s) = \frac{T_{zz}}{c_s^2} \quad (9.15)$$

Summarizing,

$$\frac{d}{dz} \begin{pmatrix} U \\ T_{zz} \end{pmatrix} = \begin{pmatrix} \frac{k^2 g_0}{\omega^2} & \frac{1}{C} - \frac{k^2}{\rho \omega^2} \\ -\rho \omega^2 + \frac{k^2 \rho g_0^2}{\omega^2} & -\frac{k^2 g_0}{\omega^2} \end{pmatrix} \begin{pmatrix} U \\ T_{zz} \end{pmatrix} \quad (9.16)$$

where z represents altitude, U is vertical displacement, and T_{zz} is stress. Here C is the bulk modulus, which is related to the sound speed c_s , density ρ , pressure p , and specific heat ratio γ by

$$C = \rho c_s^2 = \rho \left(\frac{\gamma p}{\rho} \right). \quad (9.17)$$

By simply taking the pressure as $p = -T_{zz}$, it may seem to match the hydrodynamic formulation. However, since the pressure perturbation is also taken as Eulerian in the fluid case, to compare with hydrodynamics results, it is necessary to consider the change in hydrostatic pressure due to deformation as $p = -T_{zz} + \rho_0 g_0 U$ ^{note 3)}.

9.2.1 Ocean gravity waves

^{note 3)}In elastic bodies, hydrostatic pressure arises from a decrease in timescale longer than viscous relaxation. Since the deformation of seismic waves is sufficiently shorter than the viscous relaxation timescale, the hydrostatic pressure preserves its pre-deformation value (can be considered frozen). It is possible to consistently handle it as Eulerian, but conventionally in seismology, only stress is often handled in a Lagrangian manner.

For simplification, density is approximated as a constant. Also, since the speed of sound is sufficiently faster than the speed of oceanic gravity waves, we set $1/C = 0$. The eigenvalues of the matrix are $\pm k$, and the corresponding eigenvectors are

$$\begin{pmatrix} 1 \\ \rho_0 g_0 \left(1 - \frac{\omega}{kg_0}\right) \end{pmatrix}, \begin{pmatrix} 1 \\ \rho_0 g_0 \left(1 + \frac{\omega}{kg_0}\right) \end{pmatrix}, \quad (9.18)$$

As for the solution, it must satisfy the boundary conditions $U = 0$ at the seabed and $T_{zz} = 0$ at the water surface (because stress is treated in a Lagrangian manner). Letting the water depth be H , from the boundary conditions we get

$$\tanh kH = \frac{\omega^2}{kg_0}, \quad (9.19)$$

This equation signifies the dispersion relation for oceanic gravity waves.

9.2.2 Waves in the atmosphere

When considering that the atmosphere has reached hydrostatic equilibrium, the vertical pressure gradient and gravity balance. As a result, pressure and density attenuate exponentially. Let H_s be defined as the scale height, which is the altitude interval over which the pressure drops to $1/e$. The scale height is an important physical quantity characterizing the atmospheric structure. In this section, let us consider an atmospheric structure that is density-stratified.

To be consistent with the description in meteorology (fluid dynamics), the stress is rewritten as pressure p viewed from an Eulerian perspective, and the vertical displacement U is rewritten as vertical velocity w .

$$P = \sqrt{\rho}(-T_{zz} + \rho g U) \quad (9.20)$$

$$W = \frac{i\omega U}{\sqrt{\rho}}. \quad (9.21)$$

Changing variables and rearranging the equation,

$$\frac{d}{dz} \mathbf{x}(z) = \mathbf{A}(z) \mathbf{x}(z), \quad (9.22)$$

where,

$$\mathbf{A}(z) = \begin{pmatrix} \frac{1}{2} \left(\frac{g}{c^2} - \frac{N^2}{g} \right) & -\frac{k^2}{i\omega} \frac{\omega^2 - L(k)^2}{L(k)^2} \\ \frac{N^2 - \omega^2}{i\omega} & -\frac{1}{2} \left(\frac{g}{c^2} - \frac{N^2}{g} \right) \end{pmatrix}, \quad (9.23)$$

$$\mathbf{x}(z) = \begin{pmatrix} P(z) \\ W(z) \end{pmatrix}. \quad (9.24)$$

There are two important frequencies characterizing waves in the atmosphere: the buoyancy frequency (Brunt-Väisälä frequency) N and the acoustic cut-off frequency N_a , which satisfy the following relations:

$$L(k)^2 \equiv k^2 c^2, \quad (9.25)$$

$$N^2 \equiv \frac{g}{H_\rho} - \frac{g^2}{c^2}, \quad (9.26)$$

$$N_a \equiv \frac{c}{2H_\rho}, \quad (9.27)$$

where H_ρ is a quantity called the density scale height, defined by

$$H_\rho = -\frac{dz}{d \ln \rho}. \quad (9.28)$$

There are two frequencies characterizing waves in the atmosphere: the buoyancy frequency (Brunt-Väisälä frequency) N , and the cut-off frequency N_a , which are defined as follows:

$$\begin{aligned} N^2 &= -\frac{g}{\rho_0} \frac{d\rho_0}{dz} - \frac{g^2}{c_s^2}, \\ N_a &= \frac{c_s}{2H_s}, \end{aligned} \quad (9.29)$$

where g is the gravitational acceleration, ρ_0 is the density, z is the altitude, and c_s is the speed of sound.

First, let us consider the buoyancy frequency N . Consider a small volume element. Due to density stratification, buoyancy occurs. Negative buoyancy acts when moving upwards, and positive buoyancy acts when moving downwards. As a result, oscillation takes place. This frequency is the buoyancy frequency.

Next, let us consider the cut-off frequency N_a . N_a is the lowest frequency limit at which sound waves can exist. Below this frequency, sound waves cannot exist. As can be seen from the definition, the wavelength of the sound wave at frequency N_a is $4\pi H_s$. That means the wavelength of the sound wave becomes sufficiently thicker than the atmospheric thickness H_s , so it can no longer be supported as a sound wave. N_a is always higher than the buoyancy frequency N . Elaborating broadly on the actual atmosphere, though it varies with altitude, the buoyancy frequency is roughly around 400 seconds, and the cut-off frequency is around 300 seconds.

Here, for simplicity, we consider an isothermal atmosphere. In the case of an isothermal atmosphere, the speed of sound is constant, meaning the scale height is constant, so we can write $x(z) \sim e^{\lambda z}$. Thus, the characteristic equation is,

$$\det(A - \lambda I) = \lambda^2 + \frac{f(\omega^2)}{c^2 \omega^2} = 0, \quad (9.30)$$

where f is defined as

$$f(\omega^2) = \omega^4 - (L(k)^2 + N_a^2)\omega^2 + L(k)^2 N^2. \quad (9.31)$$

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Here, let two characteristic frequencies ω_+^2 ω_-^2 ($\omega_+^2 > \omega_-^2$) be the solutions to the characteristic equation satisfying the relation:

$$f(\omega^2) = 0. \quad (9.32)$$

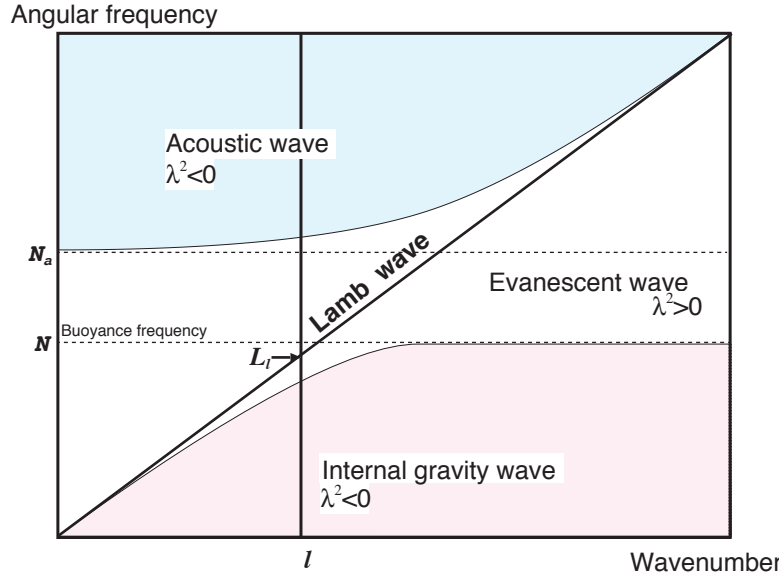


Figure 9.6: Dispersion relation of atmospheric gravity waves and sound waves. Two frequencies ω_+ and ω_- characterizing atmospheric gravity waves and sound waves act as a divide between regions where propagation as a wave in the vertical direction is possible, and the evanescent region where it cannot.

Figure 9.6 shows the dispersion relation of atmospheric gravity waves and sound waves. The two frequencies ω_+ and ω_- classifying atmospheric gravity waves and sound waves allow us to divide the propagation modes. The region where they can propagate vertically as sound waves is indicated in blue, and the region where they can propagate vertically as gravity waves in red. The white region ($\omega_- < \omega < \omega_+$) corresponds to the evanescent region^{note 4)} where λ^2 is positive, and decays exponentially in the vertical direction. The slanted line corresponds to the Lamb wave. As explained in the previous section, the Lamb wave acts as a sound wave horizontally and reaches hydrostatic equilibrium vertically. In other words, it is a boundary wave located only near the lower end of the atmosphere, with its energy attenuating proportionally to $\exp(-z/H_s)$ ^{note 5)}. The blue and red regions correspond to internal waves^{note 6)}. Because λ^2 is negative, it can propagate in the vertical direction. The acoustic wave region (blue) satisfies the condition $\omega > \omega_+$. Also, the

^{note 4)}In seismology, it is called an inhomogeneous wave, and in the atmosphere and ocean, it is also called an external wave, but physically refers to the same phenomenon.

^{note 5)}Lamb waves have been observed during total solar eclipses, atmospheric nuclear tests, and volcanic eruptions. Multi-orbit Lamb waves travelling numerous times around the Earth have also been detected. Recently, consistently excited Lamb waves believed to be sourced by atmospheric phenomena have also been discovered.⁽⁸⁾

^{note 6)}Corresponds to body waves in seismology.

acoustic wave region (blue) always exists on the shorter-period side of the cut-off frequency N_a . The other region, the internal gravity wave region (red), satisfies the relation $\omega < \omega_-$. This region consistently exists on the lower-frequency side of the buoyancy frequency N . In this way, the defining frequencies N_a , $L(k)$, and N play an important role in understanding wave propagation behaviors in the ocean and atmosphere.

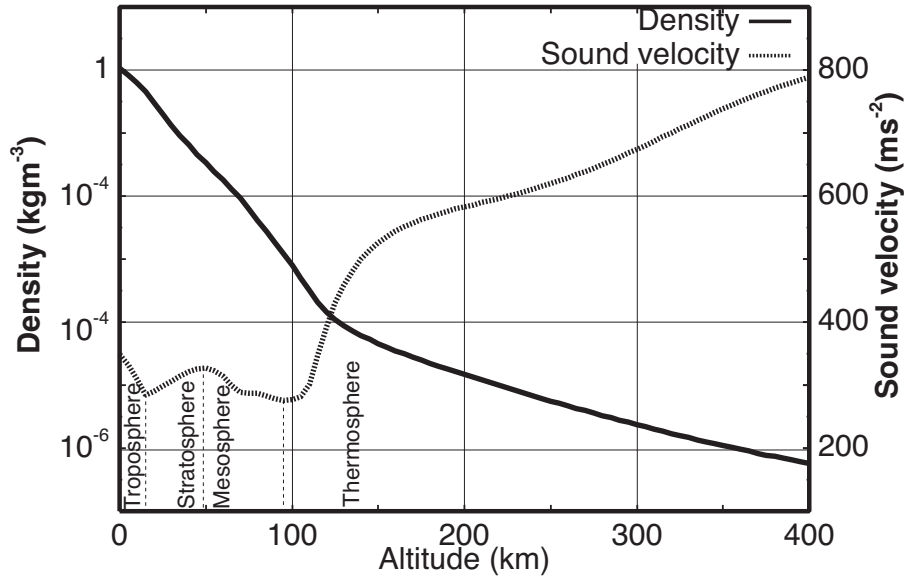


Figure 9.7: Atmospheric sound speed and density structure. Displaying the model in June for the equatorial region from CIRA86.⁽²⁾

Assuming the high-frequency limit constraint for an acoustic wave, λ^2 ($L_l N \ll \omega^2$ and $N_a \ll \omega$) can be approximated as

$$\lambda^2 = -\frac{1}{c^2}(\omega^2 - L(k)^2). \quad (9.33)$$

In this case, ω_+ can be estimated as

$$\omega_+ \sim L(k) \propto v. \quad (9.34)$$

From this, it can be understood that when a high-frequency sound wave enters a high-sound speed (high-temperature) region like the thermosphere, it will reflect.

Here, let us reflect critically upon the behavior of sound waves at low frequencies. ω_+ can be evaluated by the cut-off frequency N_a . Therefore,

$$\omega_+ \sim N_a \sim \frac{v}{H_\rho} \propto \frac{\sqrt{T}}{T} \propto v^{-1}. \quad (9.35)$$

In other words, it is found that sound waves reflect in low-temperature regions, because as evident from the equation, the vertical wavelength of the acoustic wave is blocked by the atmospheric thickness (scale height H_ρ), while concurrently the scale height intensifies linearly with temperature.

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Figure 9.8 depicts the altitude distribution of $\omega_+/2\pi$ formulated based on the COSPAR INTERNATIONAL REFERENCE ATMOSPHERE: 1986 (CIRA86) model (Figure 9.7). Comparing Figure 9.8 with Figure 9.7, we understand the relationship between ν and ω_+ . From this configuration we verify that an mHz-band acoustic wave will reflect upon joining the lower sound-speed sections of the mesosphere. This occurrence is referred to as inverted refraction. For this reason, fundamental acoustic modes and 1st higher modes are trapped in an elevation window from the surface ascending accurately to circa 100 km. The atmospheric mode trapping operates extensively shaped locally via inversed deflection (inverted refraction) through underlying low-sound gradients spanning the mesosphere. Modes beyond the 2nd higher mode barely trap because they predominantly progress extending fully over heights beyond 100 km dispersing effectively. Vertical wavenumbers dictate dynamically relying exclusively upon metrics gauging lengths separating surface to the upper reflective limits. Inherently yielding shorter footprints than horizontal dimensions uniformly, horizontal propagation group speeds belonging securely to these trapped configurations function very slowly (owing fully to repetitively reflecting vertically and dynamically traveling horizontally) (at estimations nearing 10 m/s orders).

Wavenumber-Frequency Spectrum

As shown in Figure 9.6, it is standard practice to consider wave propagation in the wavenumber-frequency domain. Especially when dispersion is strong, waveforms in the time domain become complex, often making interpretation difficult. Mathematically, considering it in the time-distance domain and considering it in the wavenumber-frequency domain are essentially equivalent. Throughout this material, various plotting methods will appear. It might be quite difficult if you are not used to them, but since there are standard plotting methods depending on the field, with a little training you will come to understand the key points of the figures.

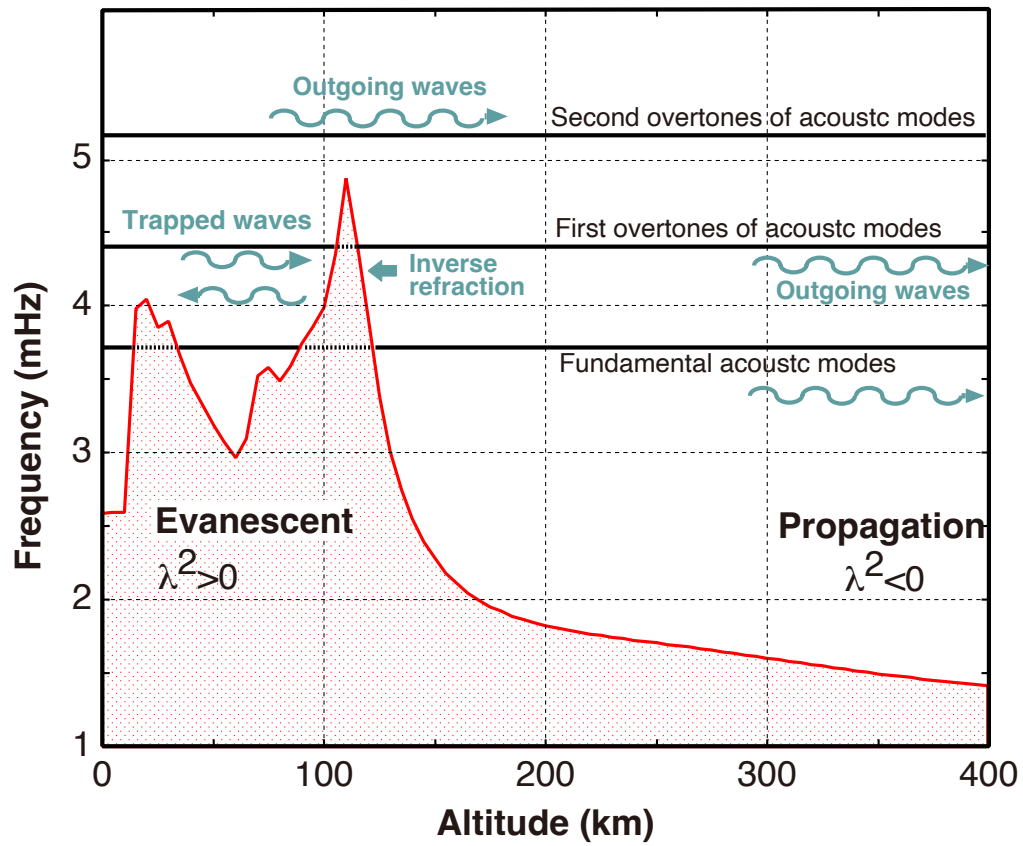


Figure 9.8: Distribution over elevation of characteristic frequency $\omega_+/2\pi$. Computations were formulated utilizing the June variables from COSPAR INTERNATIONAL REFERENCE ATMOSPHERE: 1986 (CIRA86) models precisely assessing $l = 29$ regarding the horizontal metric derived fundamentally within limits defining spherical harmonics.

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